Tail Rotor Dynamics for an RC Helicopter

Thrust, Induced Inflow, and Inner-Loop Inversion

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Abstract

This note develops a control-oriented model of an RC helicopter tail rotor. Starting from blade-element momentum theory (BEMT), we derive the thrust-versus-pitch relation, add dynamic inflow via the Pitt–Peters first-order model, and invert the coupled system to obtain the blade pitch required to track a commanded thrust. The inverse is collected into a compact implementation-ready form for the inner control loop. A linearisation about hover trim is also developed, exposing the lead–lag structure of the plant and quantifying the inflow-softening factor that varies across the operating envelope. Realistic parameter ranges for RC tail rotors are tabulated.

Contents

Nomenclature

The notation used throughout the note is collected below. Overbars denote trim (equilibrium) values; $\delta(\cdot)$ denotes small perturbations about trim; a superscript "cmd" denotes a commanded (reference) quantity. SI units are used unless otherwise stated; angles are in radians except where a table lists degrees.

Rotor geometry and operating condition

Symbol	Description	Units
R_t	Tail-rotor radius	m
r	Radial coordinate of a blade element, $0 \le r \le R_t$	m
c	Blade chord (assumed rectangular blade)	m
N_b	Number of blades	–
A_t	Rotor-disc area, $A_t = \pi R_t^2$	m^2
σ	Rotor solidity, $\sigma = N_b c/(\pi R_t)$	–
Ω_t	Rotor angular speed	$\mathrm{rad\,s^{-1}}$
$\Omega_t R_t$	Blade tip speed	$\mathrm{m\,s^{-1}}$
ρ	Air density (taken as $1.225\,\mathrm{kg\,m^{-3}}$)	$\mathrm{kg\,m^{-3}}$
Re	Blade-section Reynolds number	–

Blade-element aerodynamics

Symbol	Description	Units
θ_t	Blade collective pitch angle (control input)	rad
U_T	Tangential component of local flow at a blade section	$\mathrm{m\,s^{-1}}$
U_P	Perpendicular (axial) component of local flow	$\mathrm{m\,s^{-1}}$
ϕ	Local inflow angle, $\phi \approx U_P/U_T$	rad
α	Effective angle of attack, $\alpha = \theta_t - \phi$	rad
a	Two-dimensional lift-curve slope ($\approx 5.7\,\mathrm{rad^{-1}}$)	$\mathrm{rad^{-1}}$
C_ℓ	Section lift coefficient, $C_\ell = a\,\alpha$	–

Inflow and free-stream velocities

Symbol	Description	Units
v_i	Uniform induced velocity through the rotor disc (state)	$\mathrm{m\,s^{-1}}$
$v_{i,\mathrm{ss}}$	Steady-state value of v_i for the current operating point	$\mathrm{m\,s^{-1}}$
v_h	Hover induced velocity, $v_h = \sqrt{T_t/(2\rho A_t)}$	$\mathrm{m\,s^{-1}}$
V_∞	Axial component of the free-stream velocity through the rotor	$\mathrm{m\,s^{-1}}$
$V_\perp$	In-plane component of the free-stream velocity	$\mathrm{m\,s^{-1}}$
μ_z	Non-dimensional axial inflow, $\mu_z = V_\infty/(\Omega_t R_t)$	–
μ	Advance ratio (in-plane), $\mu = V_\perp/(\Omega_t R_t)$	–

Thrust and non-dimensional coefficients

Symbol	Description	Units
T_t	Tail-rotor thrust	N
C_T	Thrust coefficient, $C_T = T_t/[\rho A_t(\Omega_t R_t)^2]$	–
λ	Total inflow ratio, $\lambda = (v_i + V_\infty)/(\Omega_t R_t)$	–
λ_i	Induced-inflow ratio, $\lambda_i = v_i/(\Omega_t R_t)$	–
λ_h	Hover inflow ratio, $\lambda_h = \sqrt{C_T/2}$	–
λ_0	Uniform component of inflow in three-state Pitt–Peters	–
λ_s, λ_c	Longitudinal/lateral harmonic inflow components	–
C_{Mx}, C_{My}	Roll and pitch aerodynamic-moment coefficients on the disc	–

Dynamic inflow model

Symbol	Description	Units
τ_i	First-order inflow time constant (Pitt–Peters)	s
0.849	Apparent-mass coefficient, $8/(3\pi)$, in τ_i	–
$[M]$	Apparent-mass matrix in three-state Pitt–Peters	–
$[L]$	Inflow-gain matrix in three-state Pitt–Peters	–

Trim (equilibrium) quantities

Symbol	Description	Units
$\bar{T}$	Trim thrust (typically hover)	N
$\bar{v}_i$	Trim induced velocity	$\mathrm{m\,s^{-1}}$
$\bar{\theta}$	Trim blade pitch	rad
$\bar{\lambda}$	Trim inflow ratio	–
$\bar{C}_T$	Trim thrust coefficient	–
$\bar{\tau}_i$	Trim inflow time constant	s

Linearisation, plant model and compensator

Symbol	Description	Units
$\delta(\cdot)$	Small perturbation of a quantity about its trim value	(varies)
s	Laplace variable	$\mathrm{s^{-1}}$
t	Time	s
K_1	Pitch-to-thrust gain, eq. (16)	$\mathrm{N\,rad^{-1}}$
K_2	Inflow-to-thrust gain, eq. (16)	$\mathrm{N\,s\,m^{-1}}$
k_v	Linearised sensitivity $\partial v_{i,\mathrm{ss}}/\partial T$, eq. (37)	$\mathrm{m\,s^{-1}\,N^{-1}}$
$K_2 k_v$	Dimensionless DC inflow-softening coupling	–
β	Softening factor, $\beta = 1 + K_2 k_v$	–
$S(\lambda)$	Self-consistent BEMT softening factor, eq. (53)	–
x	Internal state of the lead compensator	rad

Commanded (reference) quantities

Symbol	Description	Units
T_t^{cmd}	Commanded tail-rotor thrust	N

Acronyms

BEMT	Blade Element Momentum Theory
DC	Direct current (steady-state, low-frequency limit)
TF	Transfer function
RC	Radio-controlled

1 Introduction

The tail rotor of a single-main-rotor helicopter is, in steady flight, a near-quasi-steady thrust generator: its job is to balance the anti-torque of the main rotor and, via the pilot's pedal input, to provide directional control. On a radio-controlled (RC) model helicopter the same physics applies, but the rotor operates at much lower Reynolds number, much higher rotational speed, and with a single collective control. From a flight-control perspective, the relevant question is how the commanded blade pitch θ_t maps to the thrust T_t actually produced, in both steady and transient conditions.

Two effects make this map non-trivial. First, the thrust produced at a given pitch depends on the induced velocity v_i through the rotor: as the rotor generates lift, it accelerates air downwards through the disc, which changes the local angle of attack of every blade element and reduces the effective incidence. Second, the induced velocity itself takes a finite time to develop, because the wake has inertia; the relationship between v_i and T_t is therefore dynamic, not algebraic. The combined effect is a *lead–lag* response in thrust to a step in pitch: an immediate transient at the open-loop (no-inflow) gain, followed by partial relaxation to a lower steady value as inflow builds up.

This note develops a compact, control-oriented model of these effects. Section 1 derives the blade-element thrust relation, an expression linear in pitch and in induced velocity. Section 2 closes the loop on v_i using momentum theory and the first-order Pitt–Peters dynamic-inflow model. Section 3 inverts the coupled system to obtain the pitch required to track a thrust command. Section 4 linearises the same plant about hover trim, exposing its lead–lag structure in the frequency domain and quantifying the *inflow-softening factor* that varies across the operating envelope. Section 5 collects the inverse into a compact form suitable for direct implementation in an inner control loop. Section 6 tabulates parameter ranges for typical RC-helicopter classes and works a representative numerical example.

2 Blade Pitch and Rotor Thrust

The Introduction set out the central problem: predict the thrust T_t that a tail rotor produces given the pitch θ_t applied to its blades. This chapter builds the first half of that prediction—an expression for thrust from the geometry and kinematics of a single rotating blade, at a frozen instant in time, conditional on the induced velocity v_i through the disc. The second half (computing v_i itself) is taken up in Section 3.

The tool is *Blade Element Theory* (BET), which decomposes the rotor into thin radial strips, treats each strip as a two-dimensional aerofoil section, and integrates the resulting elemental lift across span and around the disc. Combined with momentum theory in Section 3 it becomes *Blade Element Momentum Theory* (BEMT). The blade-element half is purely kinematic and aerodynamic; it does not invoke any global mass or momentum balance. That is what makes it tractable as a standalone derivation.

2.1 The Rotor and the Blade-Element Abstraction

We model the tail rotor as N_b identical blades of length R_t and chord c, rotating at constant angular speed Ω_t about a shaft, sweeping out a disc of area $A_t = \pi R_t^2$. For an RC tail rotor N_b is typically 2; for full-scale tail rotors it may be 3 or 4. The blades are set to a single *collective* pitch θ_t common to all blades and to every azimuthal position. This is a meaningful simplification specific to tail rotors: main rotors additionally vary their pitch sinusoidally with azimuth (cyclic pitch) to control the disc's tilt, but tail rotors normally do not—they have only one degree of freedom, collective pitch, which acts as the thrust command.

The blade-element idea is to chop each blade into thin radial strips of width dr at radial station r, treat each strip as an isolated two-dimensional aerofoil section that sees its own local apparent wind, compute the elemental lift and drag on the strip from 2D aerofoil theory, then sum (integrate) over the span and multiply by N_b. The abstraction works because the spanwise extent of a strip is large compared with the chord: lift on any given strip is dominated by 2D mechanisms and depends only weakly on the flow at neighbouring spanwise stations. The cost is that 3D effects—tip vortices, trailing-edge wake roll-up, root flow distortion—are excluded by construction and must be patched back in afterwards if needed (see the assumptions list at the end of the chapter).

2.2 Velocity Components at the Blade Element

At radial station r on the blade, two velocity components of the apparent wind matter. The first is *tangential*, lying in the plane of the disc and pointing opposite to the blade's motion:

$$U_T \;=\; \Omega_t\, r, \tag{1}$$

zero at the hub, linear in r, peaking at $\Omega_t R_t$ at the tip. This is what the blade sees just from rotating: the element is moving forward through still air at speed $\Omega_t r$, so the air appears to move backward past it at the same speed.

The second is *perpendicular*, normal to the disc plane, pointing through the disc:

$$U_P \;=\; v_i \;+\; V_\infty. \tag{2}$$

v_i is the induced velocity produced by the rotor itself (present whenever the rotor produces thrust, drawn through the disc by the pressure jump the blades create), and V_∞ is any external axial component of the freestream—for a tail rotor on a translating airframe, this is the sideways translation along the rotor shaft. In hover, $V_\infty = 0$ and the perpendicular flow is purely the rotor's own induced velocity. The two notations V_∞ and $\mu_z \Omega_t R_t$ refer to the same quantity, with μ_z the non-dimensional form used in some textbooks.

The velocity triangle. Imagine looking edge-on at the blade element from outboard, so its chord is perpendicular to the page. The element moves forward in the disc plane at speed U_T, and at the same time air is being drawn through the disc at speed U_P. In the element's frame these combine into an apparent wind of magnitude $U = \sqrt{U_T^2 + U_P^2}$ blowing toward the element at an angle

$$\phi \;=\; \tan^{-1}\!\left(\frac{U_P}{U_T}\right) \;\approx\; \frac{U_P}{U_T} \tag{3}$$

below the disc plane. ϕ is the *inflow angle*: the amount by which the apparent wind is tilted out of the plane in which the blade moves. Away from the root—where U_T is small and the small-angle approximation fails—typical values are $U_P \sim$ a few m/s against $U_T \sim$ tens of m/s, so $\phi \sim 0.05$–$0.15\,\mathrm{rad}$. Small-angle is justified across most of the span.

2.3 Effective Angle of Attack and Elemental Thrust

The blade is set at geometric pitch θ_t relative to the disc plane. But the section aerofoil cares only about its angle to the *apparent wind*, not to the disc. Since the apparent wind is tilted down by ϕ, the angle that the chord line makes with the flow—the *effective angle of attack*—is reduced:

$$\alpha \;=\; \theta_t \;-\; \phi. \tag{4}$$

This is the central kinematic mechanism of the chapter. The geometric pitch θ_t is what we command via the servo, but the element only feels α, and induced inflow—through ϕ—steals incidence from the blade. Every newton of thrust the rotor produces makes its own future thrust slightly harder to produce. This is what the integrated result will quantify, and what every subsequent chapter must contend with.

From angle of attack to elemental lift. Assume a linear two-dimensional lift curve, $C_\ell = a\,\alpha$, with slope a. For an inviscid thin symmetric aerofoil the slope is 2π per radian ($\approx 6.28\,\mathrm{rad}^{-1}$); real aerofoils at typical RC tail rotor Reynolds numbers give $a \approx 5.5$–$5.9\,\mathrm{rad}^{-1}$, and we adopt $a \approx 5.7\,\mathrm{rad}^{-1}$ as a representative value. The elemental lift on a strip of width dr at station r is then

$$dL \;=\; \tfrac{1}{2}\rho U^2\, c\, C_\ell\, dr \;=\; \tfrac{1}{2}\rho\,(U_T^2 + U_P^2)\, c\, a\, \alpha\, dr, \tag{5}$$

acting perpendicular to the apparent wind.

From elemental lift to elemental thrust. Thrust on the rotor is the component of aerodynamic force along the rotor axis, i.e. perpendicular to the disc plane. The lift vector sits perpendicular to the apparent wind, which is itself tilted by ϕ below the disc plane, so the lift is tilted by ϕ *away from* the rotor axis. There is also a drag vector aligned with the apparent wind, contributing in the opposite sense. The exact projection is

$$dT \;=\; dL\cos\phi \;-\; dD\sin\phi. \tag{6}$$

For small ϕ, $\cos\phi \approx 1$ (the lift projects almost entirely onto the rotor axis), and $\sin\phi \approx \phi$ so the drag term is of order $\phi\, C_d/C_\ell$ times the lift term—small, since ϕ is small and C_d is much less than C_ℓ at typical angles of attack. Dropping both corrections, and additionally neglecting U_P^2 against U_T^2 in the apparent-wind magnitude (consistent with ϕ small), we obtain

$$dT \;\approx\; \tfrac{1}{2}\rho\,(\Omega_t r)^2\, c\, a\,(\theta_t - \phi)\, dr. \tag{7}$$

Each small-angle simplification is itself $O(\phi^2)$; collectively they limit the validity of the derivation to a band of mostly small inflow angles.

2.4 Integration Across the Span

Total rotor thrust is the elemental thrust (7) summed across one blade's span and across all N_b blades:

$$T_t \;=\; N_b \int_0^{R_t} dT \;=\; \tfrac{1}{2}\rho\, a\, c\, N_b\, \Omega_t^2 \int_0^{R_t} (\theta_t - \phi)\, r^2\, dr. \tag{8}$$

Assume the induced velocity is *uniform* across the disc—i.e. v_i does not vary with r or with azimuth. This is the standard BEMT simplification. Under it, U_P is constant in r, and $\phi(r) = U_P/(\Omega_t r) = (v_i + V_\infty)/(\Omega_t r)$ varies with r only through its denominator. The integrand splits into two pieces:

$$\int_0^{R_t} \theta_t\, r^2\, dr \;=\; \frac{\theta_t\, R_t^3}{3}, \qquad \int_0^{R_t} \phi(r)\, r^2\, dr \;=\; \frac{v_i + V_\infty}{\Omega_t}\,\frac{R_t^2}{2}. \tag{9}$$

Both pieces are tip-weighted—the pitch contribution as r^2 and the inflow loss as r—because elemental thrust scales with dynamic pressure, which goes as $(\Omega_t r)^2$. Outboard stations do most of the work; inboard stations are nearly irrelevant. This is the reason rotors are conventionally analysed in terms of tip quantities.

2.5 Non-Dimensional Form

Substituting the two integrals back into the thrust expression and collecting:

$$T_t \;=\; \tfrac{1}{2}\rho\, a\, c\, N_b\, \Omega_t^2 \left[\frac{\theta_t\, R_t^3}{3} - \frac{v_i + V_\infty}{\Omega_t}\,\frac{R_t^2}{2}\right]. \tag{10}$$

Divide both sides by $\rho\, A_t\, (\Omega_t R_t)^2$, the natural dynamic scale (air density times disc area times tip-speed squared). Two non-dimensional groups emerge:

$$\sigma \;\equiv\; \frac{N_b\, c}{\pi R_t}, \qquad \lambda \;\equiv\; \frac{v_i + V_\infty}{\Omega_t R_t}. \tag{11}$$

σ is the *rotor solidity*: the fraction of the disc's swept area that is actually covered by blade planform. It is "how much rotor there is per unit disc area." Typical RC tail rotors have $\sigma \approx 0.10$–0.20, somewhat higher than main rotors ($\sigma \approx 0.05$–0.10) because tail rotors are short-radius and need proportionally more blade chord to generate their share of thrust. λ is the *inflow ratio*: the through-disc velocity expressed as a fraction of tip speed. It is small (~ 0.05–0.15 in hover) because the tip moves much faster than the wake.

The thrust coefficient

$$C_T \;\equiv\; \frac{T_t}{\rho\, A_t\, (\Omega_t R_t)^2} \tag{12}$$

is thrust non-dimensionalised by the same scale—a kind of "thrust per unit tip dynamic pressure per unit disc area." Substituting these groups gives the standard BEMT result:

$$\boxed{C_T \;=\; \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right).} \tag{13}$$

The factor $1/3$ on θ_t is the r^2 weighting of the pitch contribution; the factor $1/2$ on λ is the r weighting of the inflow loss. Dimensional thrust is recovered as

$$T_t \;=\; C_T\, \rho\, A_t\, (\Omega_t R_t)^2. \tag{14}$$

2.6 Dimensional Form: The Coefficients K_1 and K_2

The BEMT result (13) is exactly linear in θ_t and λ (and hence linear in θ_t and v_i at fixed V_∞, and linear in θ_t and v_i and V_∞ together). For control work it is convenient to absorb all the dimensional and aerodynamic constants into two coefficients, leaving a clean linear map from pitch and induced velocity to thrust:

$$\boxed{T_t \;=\; K_1\,\theta_t \;-\; K_2\,v_i \;-\; K_2\,V_\infty,} \tag{15}$$

with

$$K_1 \;=\; \frac{\rho\,A_t\,(\Omega_t R_t)^2\,\sigma\,a}{6}, \qquad K_2 \;=\; \frac{\rho\,A_t\,\Omega_t R_t\,\sigma\,a}{4}. \tag{16}$$

At fixed Ω_t (held by the main-rotor governor on an RC helicopter), K_1 and K_2 are constants of the rotor; they depend on geometry (R_t, c, N_b), air density, lift-curve slope, and tip speed, but not on the operating point (θ_t, v_i). *Equation (15) is exact within the BEMT assumptions listed below.* It is not a linearisation about any particular point; the linearity is a property of the underlying derivation.

Physical interpretation of K_1 and K_2. K_1 is the thrust the rotor would produce per radian of collective if no induced inflow developed: the open-loop or "frozen-wake" pitch effectiveness. It is what the rotor delivers in the first instant after a step command, before the wake has had time to react. K_2 is the rate at which thrust is given back as the wake builds: how much thrust each m/s of induced velocity costs. Both coefficients share the common factor $\rho A_t \sigma a$—air, geometry, aerofoil—but they differ in their tip-speed dependence. K_1 carries $(\Omega_t R_t)^2$ (it is dynamic-pressure-like), while K_2 carries only $\Omega_t R_t$ (it is velocity-like). Their ratio is therefore purely kinematic:

$$\frac{K_2}{K_1} \;=\; \frac{3}{2\,\Omega_t R_t}. \tag{17}$$

The ratio is independent of blade geometry, density, lift-curve slope, or solidity—it depends only on tip speed. Faster-spinning rotors are less penalised by inflow, in the precise sense that K_2/K_1 is smaller. The inflow-softening factor that emerges in Section 5.5 inherits this scaling.

2.7 Assumptions and Where Each Is Repaired

Equation (15), and the BEMT result (13) from which it descends, rest on the following simplifications:

- *Uniform induced velocity v_i across the disc.* Repaired in Section 3 via momentum theory, which *supplies* the value of v_i given the thrust. The one-state Pitt–Peters extension makes v_i dynamic; the three-state version restores a degree of spanwise and azimuthal non-uniformity.
- *Small inflow angle,* $\phi \ll 1$, justifying $\tan\phi \approx \phi$, $\cos\phi \approx 1$, and $U_P^2 \ll U_T^2$. Holds across most of the span but breaks down near the root, where U_T is small; root cutout mitigates this.
- *Linear two-dimensional lift curve,* $C_\ell = a\alpha$, up to stall. Stall, Reynolds-number effects, and Mach-number effects (less relevant for RC) require empirical or table-lookup corrections to a.
- *Rectangular, untwisted blade with constant chord c.* Non-rectangular or twisted blades introduce r-dependent factors inside the span integrals but no change to the basic structure.
- *No tip loss and no root cutout* (integration from 0 to R_t). Tip loss is conventionally handled by multiplying C_T by a Prandtl tip-loss factor $B \approx 0.97$; root cutout by truncating the lower integration limit.
- *Thrust contribution from section drag is ignored* ($O(\phi^2)$, see Section 2.3).

- *Constant rotor speed* Ω_t, held by the governor.

Section 7 quantifies where each of these breaks down for RC-scale tail rotors. The first item—uniform inflow—is the load-bearing one for what follows: the rest of the document is, in a sense, a careful study of what happens once v_i is allowed its own dynamics.

3 Induced Inflow: Momentum Theory

The blade-element derivation of Section 1 produced $C_T = (\sigma a/2)(\theta_t/3 - \lambda/2)$, which contains *two* unknowns: the pitch θ_t (which we set) and the inflow ratio λ (which we do not). To close the system we need a second relationship between thrust and induced velocity. *Momentum theory* supplies it by treating the rotor disc as a whole and applying conservation of mass and momentum to the column of air passing through it.

3.1 The Actuator Disc and the Mass-Flow Argument

Replace the physical rotor by an idealised *actuator disc*: an infinitesimally thin permeable surface of area A_t that imparts a uniform pressure jump Δp to the air passing through it, without imparting swirl or losses. Draw a streamtube enclosing the disc, extending from far upstream (station 1, where the flow is undisturbed) through the disc (station 2, just below it—or station 3 just above, depending on orientation) and out to far downstream (station 4, where pressure has returned to ambient). For a rotor in hover the upstream flow is at rest and the streamtube contracts as the air accelerates; for a rotor in axial climb the upstream flow has velocity V_∞.

Three relationships govern this column of air:

1. *Mass conservation.* The mass flow rate $\dot{m} = \rho A_t U$ is the same at every station, where U is the local axial velocity. At the disc, $U = V_\infty + v_i$, where v_i is the induced velocity *at the disc.*
2. *Momentum.* The thrust on the rotor is the reaction to the change in axial momentum of the air, evaluated between far upstream and far downstream: $T_t = \dot{m}\,(w_\infty - V_\infty)$, where w_∞ is the axial velocity in the far wake.
3. *Bernoulli, separately above and below the disc.* Energy is conserved along streamlines on each side of the disc, but *not* across it (the rotor does work on the fluid). Applying Bernoulli from station 1 to just above the disc, and from just below the disc to station 4, and using the pressure jump Δp at the disc gives, after some algebra, the celebrated *Froude result*:

$$w_\infty \;=\; V_\infty + 2\,v_i. \tag{18}$$

The Froude result is non-obvious and worth dwelling on: *half* the total velocity change is acquired upstream of the disc, and the other half downstream. The wake therefore contracts to half the disc area and moves at twice the induced velocity. This factor of two appears in every momentum-theory thrust expression and is the reason rotors are *efficient* thrust generators at low disc loading.

Substituting $w_\infty - V_\infty = 2v_i$ into the momentum equation, together with $\dot{m} = \rho A_t(V_\infty + v_i)$, gives the *axial-flight* form of the momentum balance:

$$T_t \;=\; 2\,\rho\,A_t\,v_i\,(V_\infty + v_i). \tag{19}$$

3.2 Edgewise Flow: The Full Momentum Balance

For a tail rotor mounted on a translating airframe, the free-stream is not generally aligned with the rotor axis. Decompose the airframe velocity at the rotor into a component along the rotor axis (axial, V_∞, which feeds the streamtube directly) and a component in the disc plane (edgewise, $V_\perp$, which sweeps air sideways through the disc). The resultant velocity that the disc accelerates is the vector sum of the in-plane sweep $V_\perp$ and the through-disc flow $V_\infty + v_i$:

$$U_{\text{disc}} \;=\; \sqrt{(V_\infty + v_i)^2 + V_\perp^2}. \tag{20}$$

Glauert's hypothesis takes the mass-flow rate through the disc to be $\dot{m} = \rho A_t U_{\text{disc}}$, and the axial momentum imparted to be $2v_i$ (as in the purely axial case). The momentum equation then generalises to

$$\boxed{T_t \;=\; 2\,\rho\,A_t\,v_i\,\sqrt{(V_\infty + v_i)^2 + V_\perp^2}.} \tag{21}$$

The square-root factor is just the resultant velocity at the disc; the factor of two is Froude. In the purely axial case ($V_\perp = 0$) we recover (19), and in the hover limit ($V_\infty = V_\perp = 0$) the relation simplifies further to

$$T_t \;=\; 2\,\rho\,A_t\,v_i^2 \qquad \Longrightarrow \qquad v_h \;\equiv\; \sqrt{\frac{T_t}{2\,\rho\,A_t}}. \tag{22}$$

Reading the hover result. Equation (22) is the single most-cited result of rotor aerodynamics. It says induced velocity is set by *disc loading* T_t/A_t: rotors with larger disc area need less induced velocity for the same thrust, and induced power $T_t\,v_h \propto T_t^{3/2}/\sqrt{A_t}$ is lower as a result. A tail rotor (small A_t, high T_t/A_t) therefore has *high* v_i compared with a main rotor of equal thrust—a numerical example in Section 7 gives $v_h \sim 9\,\mathrm{m\,s^{-1}}$ for a typical 700 mm-class tail rotor producing 12 N of thrust.

3.3 Non-Dimensional Form

Dividing (21) through by $\rho A_t (\Omega_t R_t)^2$ converts it to a relation between non-dimensional groups. With the inflow ratios $\lambda_i = v_i/(\Omega_t R_t)$ and $\lambda = (V_\infty + v_i)/(\Omega_t R_t)$, and the edgewise advance ratio $\mu = V_\perp/(\Omega_t R_t)$, the result is

$$C_T \;=\; 2\,\lambda_i\,\sqrt{\mu^2 + \lambda^2} \qquad \Longleftrightarrow \qquad \lambda_i \;=\; \frac{C_T}{2\,\sqrt{\mu^2 + \lambda^2}}. \tag{23}$$

The square-root $\sqrt{\mu^2 + \lambda^2}$ is the non-dimensional resultant velocity at the disc: μ is the in-plane (edgewise) component, λ the through-disc component, and their hypotenuse is what the disc actually accelerates.

Dimensional and non-dimensional forms in parallel. From here on we move between the dimensional induced velocity v_i and the inflow ratio λ_i freely; they encode the same physical quantity, related by $\lambda_i = v_i/(\Omega_t R_t)$. The dimensional form is convenient inside the dynamic-inflow ODE (Section 3.5), the non-dimensional form inside BEMT algebra, and the hover form v_h for sizing arguments.

3.4 Closing the Coupled System

Equation (23) and the BEMT result (13) are two relations between the same three quantities (θ_t, C_T, λ). For a given pitch θ_t they can be solved simultaneously for the steady-state inflow. In hover ($\mu = 0$ and $V_\infty = 0$, so $\lambda = \lambda_i$), substituting C_T from BEMT into the momentum result gives

$$\lambda_i \;=\; \frac{(\sigma a/2)(\theta_t/3 - \lambda_i/2)}{2\lambda_i} \qquad \Longrightarrow \qquad \lambda_i^2 + \frac{\sigma a}{8}\lambda_i - \frac{\sigma a}{12}\theta_t \;=\; 0. \tag{24}$$

This is a quadratic in λ_i whose positive root is the standard hover-BEMT inflow:

$$\boxed{\lambda_i \;=\; \frac{\sigma a}{16}\left[\sqrt{1 + \frac{64\,\theta_t}{3\,\sigma a}} \;-\; 1\right] \qquad \text{(hover, } \mu = 0\text{).}} \tag{25}$$

For small θ_t the bracket expands as $\frac{32\theta_t}{3\sigma a} - \mathcal{O}(\theta_t^2)$, giving $\lambda_i \approx 2\theta_t/3$—a useful order-of-magnitude check.

3.5 Wake Inertia

Equation (25) gives the *steady* inflow for a given pitch, but it ignores time. In reality the wake is a column of air with mass, and changing the rotor's thrust requires accelerating that column—which cannot happen instantaneously. If at time $t = 0$ the pilot steps the pitch up, the blade-element thrust jumps immediately to its open-loop value $K_1\,\delta\theta_t$ (the wake hasn't moved yet, so the local angle of attack is still $\theta_t - \phi$ at the old ϕ). Over the next few milliseconds the induced velocity builds toward its new steady value, the local angle of attack drops, and the thrust relaxes to its lower equilibrium. The rotor exhibits a *lead–lag* response in thrust because the inflow has *lag*.

For control-design purposes the simplest faithful model treats the uniform component of the induced velocity as a first-order state:

$$\boxed{\tau_i\,\dot{v}_i + v_i \;=\; v_{i,\mathrm{ss}},} \tag{26}$$

where $v_{i,\mathrm{ss}}$ is the steady-state induced velocity that solves (21) for the *current* thrust T_t, and τ_i is the wake-relaxation time constant. This is the *one-state Pitt–Peters* dynamic inflow model. Equation (26) is the inflow analogue of an RC low-pass filter: $v_{i,\mathrm{ss}}$ is the target, v_i is the state, and τ_i sets how quickly the state chases the target.

3.6 The Pitt–Peters Time Constant

What sets τ_i? Two mechanisms contribute. The first is the simple flow-traversal time: an air parcel takes roughly R_t/v_h to cross the rotor disc, so τ_i must scale this way on dimensional grounds alone. The second, more subtle effect is the *apparent mass* of the air column: when an impermeable disc is accelerated through fluid, the fluid in its immediate neighbourhood is accelerated with it, contributing an extra inertia even in the absence of a true rotor wake. Lamb's classical potential-flow result gives the apparent mass of a thin circular disc as $\rho\,(8/3)\,R_t^3$, a fraction $8/(3\pi) \approx 0.849$ of the mass of a sphere of radius R_t. Pitt and Peters identified this same coefficient as governing the lag of the uniform-inflow component, yielding

$$\boxed{\tau_i \;=\; \frac{8/(3\pi)}{4\,\Omega_t\,\lambda_i^{\mathrm{trim}}} \;=\; \frac{0.849\,R_t}{4\,v_h},} \tag{27}$$

where the two forms are equivalent via $\lambda_i^{\mathrm{trim}} = v_h/(\Omega_t R_t)$ at trim, and the substitution $\Omega_t\,\lambda_i^{\mathrm{trim}} R_t = v_h$ collapses the first form into the second. The right-hand form is the easier to interpret: τ_i is essentially the time for an air parcel travelling at the hover induced velocity to cross a quarter of the disc radius, scaled by the apparent-mass coefficient. Higher disc loading $\Rightarrow$ faster $v_h \Rightarrow$ shorter τ_i: hard-working rotors have snappier wakes.

Caveat on Pitt–Peters time constants. The closed-form (27) is known to underpredict measured inflow lag on full-scale rotors by a factor of 2–4. For RC tail rotors it likely underpredicts somewhat less, but the value should be regarded as a lower bound; see Section 7.

3.7 The Coupled Model

Combining the blade-element thrust relation with the dynamic-inflow state gives the complete tail-rotor model of this note:

$$T_t = \rho A_t(\Omega_t R_t)^2\,\frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{v_i + V_\infty}{2\Omega_t R_t}\right), \tag{28}$$

$$\dot{v}_i = \frac{1}{\tau_i}\left(\frac{T_t}{2\rho A_t\sqrt{(V_\infty + v_i)^2 + V_\perp^2}} - v_i\right). \tag{29}$$

Equation (28) is BEMT, algebraic in θ_t and v_i. Equation (29) is Pitt–Peters: a first-order ODE in v_i driven by the instantaneous thrust through the momentum-theory steady-state map. The pair is a one-state nonlinear system with input θ_t and output T_t, parameterised by the flight condition $(V_\infty, V_\perp)$ and the rotor constants. This pair is inverted in Section 3, analysed in the frequency domain—and characterised across the operating envelope—in Section 5, and packaged for implementation in Section 6.

3.8 Three-State Pitt–Peters (Extension)

The one-state model captures only the *uniform* component of induced velocity. In forward flight the inflow becomes non-uniform: the advancing-side blade sees different inflow from the retreating-side blade, and a longitudinal gradient develops as the wake skews backward under $V_\perp$. Pitt and Peters captured these effects with two additional states—λ_s (longitudinal inflow harmonic, tilting the inflow front-to-back) and λ_c (lateral harmonic, tilting it side-to-side)—driven by the aerodynamic pitching and rolling moments on the disc:

$$[M]\begin{Bmatrix}\dot\lambda_0\\ \dot\lambda_s\\ \dot\lambda_c\end{Bmatrix} + [L]^{-1}\begin{Bmatrix}\lambda_0\\ \lambda_s\\ \lambda_c\end{Bmatrix} = \begin{Bmatrix}C_T\\ C_{Mx}\\ C_{My}\end{Bmatrix}, \tag{30}$$

with apparent-mass matrix $[M] = \mathrm{diag}\big(\frac{8}{3\pi}, \frac{16}{45\pi}, \frac{16}{45\pi}\big)$ and gain matrix $[L]$ depending on advance ratio and wake skew. The harmonic components matter for main rotors in forward flight and for cyclic-pitch control, but for a tail rotor in near-hover, with no cyclic and only modest edgewise flow from main-rotor wake, the uniform component dominates and the one-state model (26) is almost always sufficient. We use it for the remainder of this note.

4 Inverse Problem: Blade Pitch from Commanded Thrust

The coupled model of Section 3—blade-element thrust (28) and dynamic inflow (29)—maps a pitch input θ_t to a thrust output T_t. For control design we need to run this map *backwards*: given a commanded thrust $T_t^{\rm cmd}(t)$, what pitch $\theta_t(t)$ should we apply? The answer will sit inside the tail-rotor inner loop as a feedforward block: the outer loop (yaw control) produces a thrust demand, the inverse block translates it into pitch, and the servo drives the swashplate.

This chapter derives the inverse in two steps. First, we invert the blade-element relation algebraically to obtain the *instantaneous* pitch–thrust inverse, assuming the induced velocity is known (Section 4.1). Second, we expand to the dynamic case by carrying the induced velocity as a state, governed by the Pitt–Peters ODE (Section 4.2). The result is a complete nonlinear dynamic inversion of the rotor plant. Section 5 linearises the system about hover trim, characterises its lead–lag response in the frequency domain, and quantifies the resulting inflow-softening factor across the operating envelope; Section 6 then collects the inverse into a compact form for direct implementation in an inner control loop.

4.1 The Instantaneous Inverse (Algebraic Step)

The blade-element relation (15) is linear in $(\theta_t, v_i, V_\infty)$ at fixed Ω_t, and therefore trivially invertible for θ_t:

$$\boxed{\theta_t \;=\; \frac{T_t^{\rm cmd} \;+\; K_2\big(v_i + V_\infty\big)}{K_1}.} \tag{31}$$

This is the *exact* inverse at every instant, conditional on knowing the current v_i and V_∞: apply pitch (31) and the blade element will, instantaneously, produce thrust $T_t^{\rm cmd}$.

Two terms appear on the right-hand side. The first, $T_t^{\rm cmd}/K_1$, is the pitch the rotor would need if no induced inflow were present—a naive "pitch per unit thrust" calculation. The second, $K_2(v_i + V_\infty)/K_1$, is the extra pitch needed to overcome inflow's stolen incidence: v_i and V_∞ together set the inflow angle ϕ, and the blade must be pitched up by the same amount just to retain its original effective angle of attack α. Equation (31) is, in essence, the control-theoretic restatement of the kinematic story from Section 1.

4.2 Expanding to the Dynamic System

Equation (31) requires the *current* value of v_i. On an RC platform v_i is not directly measurable: there is no sensor that reports the induced velocity through the disc. The inverse must therefore carry an internal estimate of v_i, propagated by the same physics that governs the real rotor's inflow— namely, the Pitt–Peters ODE (26) closed by the momentum relation (21). Concretely:

$$\dot{v_i} \;=\; \frac{1}{\tau_i}\big(v_{i,\rm ss} - v_i\big), \tag{32}$$

$$v_{i,\rm ss} \;=\; \frac{T_t^{\rm cmd}}{2\rho A_t\sqrt{(V_\infty + v_i)^2 + V_\perp^2}}, \tag{33}$$

$$\tau_i \;=\; \frac{0.849}{4\,\Omega_t\lambda_i}, \qquad \lambda_i = v_i/(\Omega_t R_t). \tag{34}$$

Equation (32) is the wake's relaxation: the induced velocity chases its steady-state target $v_{i,\rm ss}$ on a timescale τ_i. Equation (33) is the momentum closure: the target $v_{i,\rm ss}$ is determined by the thrust the rotor is asked to produce, with edgewise flow $V_\perp$ modifying the disc's effective mass-flow rate. Equation (34) is the apparent-mass time constant from Section 3, here written explicitly as a function of the current v_i—so τ_i itself is mildly state-dependent.

Substituting the algebraic inverse (31) for θ_t and using the dynamic state v_i from (32)–(34), we have a complete *nonlinear dynamic inversion*: a one-state nonlinear system whose input is the commanded thrust and whose output is the pitch command. No linearisation is involved; no operating point has been fixed; no small-signal assumption has been made. The model is exact within the BEMT and Pitt–Peters assumptions inherited from Sections 1–2.

A subtle point: $v_{i,\mathrm{ss}}$ in (33) appears to be driven by T_t^{cmd} (what we want), not by T_t (what we produce). At a control-theoretic level these are the same if the inverse is working—θ_t is computed precisely so that $T_t = T_t^{\mathrm{cmd}}$ at the blade element—so driving the inflow estimator from T_t^{cmd} is self-consistent and avoids needing a thrust measurement. The choice does become relevant when servo or sensor delays are introduced, but that is outside this chapter's scope.

5 Linearised Plant Analysis

The nonlinear plant (28)–(29) of Section 3—and its inverse derived in Section 4.2—can be analysed in the frequency domain by linearising about a chosen trim point. The resulting transfer function exposes the pole and zero structure of the rotor, the "lead–lag" character of its thrust response to a step in pitch, and the inflow-softening factor that quantifies how much DC gain is lost to the wake. The chapter closes by showing how this softening factor varies across the operating envelope. The linearised model is *not* an alternative controller—the inner loop will run the full nonlinear inverse, packaged for implementation in Section 6—but it gives the language in which the plant's bandwidth and stability margins are usually discussed, and is essential when an outer loop must be designed around this plant.

5.1 Linearisation About Hover Trim

Choosing trim. By *trim* we mean an equilibrium operating point of the plant: a constant pitch $\bar{\theta}$ producing a constant thrust $\bar{T}$ with a constant induced velocity $\bar{v}_i$, all satisfying both BEMT and momentum theory simultaneously. For a tail rotor on a hovering helicopter the natural choice is the trim that balances main-rotor anti-torque in still air. We specialise to hover ($V_\infty = V_\perp = 0$) because the algebra is cleanest and because the tail rotor spends most of its operating life near this condition; the non-hover case generalises by replacing $\bar{v}_i$ with the hover-equivalent $v_h = \sqrt{\bar{T}/(2\rho A_t)}$ and adjusting the trim accordingly.

Perturbation convention. Write each variable as a constant trim value plus a small perturbation:

$$\theta_t = \bar{\theta} + \delta\theta_t, \quad v_i = \bar{v}_i + \delta v_i, \quad T_t = \bar{T} + \delta T. \tag{35}$$

We retain terms linear in the δ-quantities and discard products of two perturbations (assumed small $\times$ small). The trim values themselves satisfy the steady model and cancel out of the perturbation equations, leaving a system in the δ-variables alone.

What needs linearising. The blade-element relation (15), $T_t = K_1\theta_t - K_2 v_i$, is *already linear* in its two arguments at fixed Ω_t; its perturbation form is simply

$$\delta T \;=\; K_1\,\delta\theta_t \;-\; K_2\,\delta v_i. \tag{36}$$

The nonlinearity is entirely in the momentum relation $\bar{T} = 2\rho A_t \bar{v}_i^2$, which couples thrust to induced velocity through a square. Differentiating implicitly,

$$\left.\frac{\partial T}{\partial v_i}\right|_{\text{trim}} = 4\,\rho A_t\,\bar{v}_i, \qquad \Longrightarrow \qquad \delta v_{i,\text{ss}} \;=\; \frac{1}{4\rho A_t \bar{v}_i}\,\delta T \;\equiv\; k_v\,\delta T, \tag{37}$$

where k_v is the linearised sensitivity of steady-state induced velocity to thrust at the chosen trim:

$$k_v \;\equiv\; \left.\frac{\partial v_{i,\text{ss}}}{\partial T}\right|_{\text{trim}} \;=\; \frac{1}{4\rho A_t \bar{v}_i} \;=\; \frac{\bar{v}_i}{2\bar{T}}, \qquad \text{units: } \mathrm{m\,s^{-1}\,N^{-1}}. \tag{38}$$

The two forms come from the same expression via the hover relation $\bar{T} = 2\rho A_t \bar{v}_i^2$. Physically: k_v tells us *how much extra induced velocity each newton of additional thrust will pull through the disc, in steady state.* It is large for low-thrust trims (where $\bar{v}_i$ is small and any new thrust must accelerate a slow wake from near-rest) and small for high-thrust trims (where $\bar{v}_i$ is already substantial and incremental thrust matters less proportionally).

Linearised inflow dynamics. The Pitt–Peters ODE (26) is already linear in v_i at fixed τ_i. Substituting $v_{i,\text{ss}} = \bar{v}_i + k_v\,\delta T$ and subtracting the trim equation $\bar{v}_i = \bar{v}_i$ leaves

$$\tau_i\,\delta\dot{v}_i + \delta v_i \;=\; k_v\,\delta T. \tag{39}$$

Taking the Laplace transform with zero initial conditions ($\dot{(\cdot)} \to s\,(\cdot)$):

$$(\tau_i s + 1)\,\delta v_i(s) \;=\; k_v\,\delta T(s). \tag{40}$$

5.2 The Linearised Plant Transfer Function

Equations (36) and (40) are a two-equation linear system in the three Laplace-domain quantities $\delta\theta_t(s)$, $\delta T(s)$, $\delta v_i(s)$. Eliminate δv_i: from (40),

$$\delta v_i(s) \;=\; \frac{k_v}{\tau_i s + 1}\,\delta T(s). \tag{41}$$

Substitute into (36):

$$\delta T(s) \;=\; K_1\,\delta\theta_t(s) \;-\; K_2\,\frac{k_v}{\tau_i s + 1}\,\delta T(s). \tag{42}$$

Collect δT:

$$\delta T(s)\left[1 + \frac{K_2 k_v}{\tau_i s + 1}\right] \;=\; K_1\,\delta\theta_t(s), \tag{43}$$

and rearrange to standard form,

$$\boxed{\frac{\delta T(s)}{\delta\theta_t(s)} \;=\; \frac{K_1\,(\tau_i s + 1)}{\tau_i s + (1 + K_2 k_v)}.} \tag{44}$$

This is the linearised tail-rotor plant: thrust per radian of pitch perturbation about hover trim, valid for small $\delta\theta_t$ over a bandwidth limited by the linearisation.

5.3 Reading the Plant: Lead–Lag Structure

The plant (44) has one zero and one pole, both on the negative real axis:

$$\text{zero at } s = -\frac{1}{\tau_i}, \qquad \text{pole at } s = -\frac{1 + K_2 k_v}{\tau_i}. \tag{45}$$

Because $K_2 k_v > 0$, the pole sits at a higher frequency than the zero. The high-frequency and DC limits of the gain are

$$\lim_{s\to\infty} \frac{\delta T}{\delta\theta_t} = K_1 \qquad \text{(instantaneous, “open-loop” gain)}, \tag{46}$$

$$\lim_{s\to 0} \frac{\delta T}{\delta\theta_t} = \frac{K_1}{1 + K_2 k_v} \qquad \text{(steady-state, “closed-wake” gain)}. \tag{47}$$

The DC gain is lower than the instantaneous gain by a factor $1+K_2 k_v$. A step in pitch produces an immediate thrust spike of size $K_1\,\delta\theta_t$ (before the wake responds), which then relaxes over a timescale $\tau_i/(1 + K_2 k_v)$ to the lower steady value $K_1\,\delta\theta_t/(1 + K_2 k_v)$. This is the *lead–lag* response described qualitatively in the introduction, now in closed form.

Physical reading of K_2k_v. The dimensionless product K_2k_v measures how strongly induced inflow couples back into the thrust. Tracing its factors: K_2 is how much thrust each m/s of induced velocity steals; k_v is how much extra induced velocity each newton of thrust pulls through the disc. Their product closes the loop and gives the fractional thrust loss between the instantaneous and steady-state response:

$$\frac{\text{DC gain}}{\text{instantaneous gain}} = \frac{1}{1+K_2k_v}. \tag{48}$$

For typical RC tail rotors at hover trim, K_2k_v sits in the range 0.5–1.5—meaning the DC gain is anywhere from 40% to 70% of the open-loop gain. The thrust the rotor delivers in steady state is significantly less than what its blade pitch suggests. This phenomenon is *inflow softening*, and Section 5.5 quantifies it across the operating envelope with a self-consistent closed form that supersedes the naive product $K_2 \cdot k_v$ used here as a symbolic placeholder.

5.4 Inverting the Plant: The Lead Compensator

A feedforward compensator that cancels the plant dynamics is, by definition, the inverse of the plant. Reciprocating (44):

$$\boxed{\frac{\delta\theta_t(s)}{\delta T(s)} = \frac{\tau_i s + (1+K_2k_v)}{K_1\,(\tau_i s + 1)}.} \tag{49}$$

The roles of pole and zero are now swapped: the compensator has its *zero* where the plant had its *pole*, and vice versa. The compensator is, in classical control terminology, a *lead network*: its zero sits at a lower frequency than its pole, so its phase contribution is positive across the intervening band. Cascading (49) → (44) gives the identity transfer function. This is the linearised counterpart of the nonlinear dynamic inverse derived in Section 4.2: cancellation through the wake dynamics, expressed in the frequency domain rather than as a state-update.

5.5 The Inflow-Softening Factor

The DC gain $K_1/(1+K_2k_v)$ varies with the trim point because $k_v \propto 1/\bar{v}_i \propto 1/\sqrt{\bar{T}}$. The factor $(1+K_2k_v)$ is the *inflow softening factor*—softening is strongest at low thrust.

A naive estimate computes K_2 from (16) and k_v from (37) separately and multiplies. As noted in Section 5.3, this overstates the coupling because the two factors are not independent at trim. A self-consistent value follows by closing the coupled BEMT–momentum system in hover and differentiating directly. In hover, the coupled equations are

$$C_T = \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right), \tag{BEMT}$$

$$\lambda = \sqrt{C_T/2}. \tag{Momentum}$$

Eliminating C_T:

$$\lambda^2 = \frac{\sigma a}{4}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right), \tag{50}$$

with closed form

$$\lambda(\theta_t) = \frac{\sigma a}{16}\left[\sqrt{1 + \frac{64\theta_t}{3\sigma a}} - 1\right]. \tag{51}$$

Trim sensitivity. Differentiating implicitly:

$$2\lambda\,\frac{d\lambda}{d\theta_t} = \frac{\sigma a}{4}\left(\frac{1}{3} - \frac{1}{2}\frac{d\lambda}{d\theta_t}\right) \implies \frac{d\lambda}{d\theta_t} = \frac{\sigma a}{24\lambda + 3\sigma a/2}. \tag{52}$$

Then

$$\boxed{\left.\frac{dC_T}{d\theta_t}\right|_{\text{steady}} = \frac{\sigma a}{6} \cdot \underbrace{\frac{1}{1+\dfrac{\sigma a}{16\lambda}}}_{S(\lambda)} .} \tag{53}$$

The factor $\sigma a/6$ is the instantaneous (high-frequency) gain; $S(\lambda)$ is the softening factor. Comparing with the linearised form $S = 1/(1 + K_2 k_v)$ from Section 5.3 identifies the self-consistent value of the coupling product as

$$\boxed{\left.K_2 k_v\right|_{\text{self-consistent}} = \frac{\sigma a}{16\,\lambda_h}, \qquad \lambda_h = \sqrt{C_T/2}.} \tag{54}$$

This is the expression to use in design: it bakes in the trim consistency between K_2 and k_v that a naive separate calculation would miss.

5.6 Magnitude Across the Operating Envelope

For a 700 mm-class tail rotor ($\sigma = 0.136$, $a = 5.7$, hover $\bar{C}_T = 0.0076$, $\bar{\lambda} = 0.062$):

Thrust	C_T	λ	$K_2 k_v = \sigma a/(16\lambda)$	$S(\lambda)$
25% hover	0.0019	0.031	1.58	0.39
50% hover	0.0038	0.044	1.12	0.47
Hover	0.0076	0.062	0.79	0.56
150% hover	0.0113	0.075	0.65	0.61
200% hover	0.0151	0.087	0.56	0.64

The DC gain varies by roughly 40% across the operating envelope. A fixed-gain lead compensator will mis-track by this amount at the extremes.

Time constant varies too. The Pitt–Peters time constant $\tau_i = 0.849/(4\Omega_t\lambda)$ scales as $1/\lambda \propto 1/\sqrt{C_T}$. For the same 700 mm-class rotor:

Thrust	τ_i [ms]
25% hover	6.7
Hover	3.3
200% hover	2.4

Both the gain and the time constant of the rotor depend on the trim operating point, and both must be scheduled for a fixed-coefficient controller to track across the envelope.

6 Control Loop Implementation

The instantaneous inverse (31) together with the inflow ODE (32)–(34) is a complete nonlinear dynamic inversion of the tail-rotor plant. This chapter collects those equations into a compact form suitable for direct implementation in an inner control loop, separating the fixed rotor parameters from the live signals and showing the update law in the form a practitioner would actually code. The linearised analysis of Section 5—including the envelope behaviour of the softening factor—informs how the resulting controller will behave across the thrust range, but is not a prerequisite for running the loop itself.

6.1 Parameters, Inputs, State, Outputs

The system as derived has a long list of symbols. Many of them are fixed properties of the rotor and air, computed once at startup and held constant thereafter. Separating these from the live state and commands gives a compact specification for the inner loop.

Fixed rotor parameters (computed once, constant for given hardware):

$$K_1, \quad K_2, \quad \rho A_t, \quad \Omega_t R_t, \tag{55}$$

defined in (16) and the BEMT scaling of Section 1. The apparent-mass coefficient $\kappa = 8/(3\pi) \approx 0.849$ is universal and need not be re-stored as a parameter.

Inputs (provided each control tick):

$$T_t^{\mathrm{cmd}}(t) \quad [\mathrm{N}], \qquad V_\infty(t) \quad [\mathrm{m/s}], \qquad V_\perp(t) \quad [\mathrm{m/s}]. \tag{56}$$

T_t^{cmd} comes from the yaw outer loop; V_∞ and $V_\perp$ come from the airframe's airspeed estimator (or are set to zero in near-hover applications).

State (one scalar, evolved by the inverse block):

$$v_i(t) \quad [\mathrm{m/s}]. \tag{57}$$

Outputs :

$$\theta_t(t) \quad [\text{rad, to the servo}]. \tag{58}$$

6.2 The Update Law

At each control tick of period Δt:

$$\boxed{\begin{aligned}
\theta_t &= \tfrac{1}{K_1}\big[T_t^{\mathrm{cmd}} + K_2(v_i + V_\infty)\big], && (59)\\
v_{i,\mathrm{ss}} &= \frac{T_t^{\mathrm{cmd}}}{2\rho A_t\sqrt{(V_\infty + v_i)^2 + V_\perp^2}}, && (60)\\
\tau_i &= \frac{0.849\,\Omega_t R_t}{4\,\Omega_t\,v_i} = \frac{0.849\,R_t}{4\,v_i}, && (61)\\
v_i &\leftarrow v_i + \Delta t\,\frac{v_{i,\mathrm{ss}} - v_i}{\tau_i}. && (62)
\end{aligned}}$$

This is a complete control law for the tail-rotor inner loop: four numbers in $(K_1, K_2, \rho A_t, R_t)$, three inputs in, one state held, one output out. A practitioner can transcribe (59)–(62) into firmware with no intermediate analytical steps. The square root in (60) must be safeguarded against tiny denominators when both $V_\infty + v_i$ and $V_\perp$ approach zero (e.g. at rotor spin-up); a small ϵ added inside, or a clamp on v_i from below, suffices.

6.3 Common Simplifications

Two simplifications are commonly made when fitting the inner loop into a constrained embedded environment.

Near-hover $V_\infty, V_\perp \approx 0$. The momentum closure (60) collapses to $v_{i,\text{ss}} = T_t^{\text{cmd}}/(2\rho A_t\, v_i)$, avoiding the square root and the airspeed estimate. This is sound for a tail rotor whose airframe spends most of its time in near-hover or low-speed translation; the error grows with edgewise flow and should be re-examined for operating points well outside hover.

Frozen τ_i. Hold τ_i at a representative value (computed at the dominant hover trim, or via a slow-update filter on v_i), removing the state-dependence in (61). Lag dynamics then become a fixed first-order filter, which is easier to tune and analyse. The error is small when the rotor operates near its design trim and grows at the extremes of the thrust envelope; it can be quantified using the envelope tables in Section 5.6.

Equations (59)–(62) are the deliverable of this note from a control-implementation standpoint. Concrete numerical values for the fixed rotor parameters $(K_1, K_2, \rho A_t, R_t)$, and representative magnitudes for the state v_i and the time constant τ_i across the operating envelope, are tabulated in the next chapter (Section 7).

7 Parameter Ranges for RC Tail Rotors

The tables below summarise representative tail-rotor parameters for three classes of RC helicopter, indexed by *main* rotor blade length (a customary size descriptor in the hobby). The tail rotor itself is much smaller than the main rotor in each case. The ranges span gentle hover and aggressive 3D operating points; the worked example at the end picks a representative 700-class point.

7.1 Geometric and Operating Parameters

Symbol	Quantity	420 mm	550 mm	700 mm
R_t	Tail radius [m]	0.080–0.100	0.10–0.12	0.135–0.145
c	Blade chord [m]	0.015–0.022	0.022–0.026	0.028–0.032
N_b	Number of blades	2	2	2 (occ. 3)
Ω_t	Rotor speed [$\mathrm{rad\,s^{-1}}$]	1200–1700	1000–1300	900–1100
$\Omega_t R_t$	Tip speed [m/s]	95–170	100–155	120–160
σ	Solidity	0.10–0.18	0.12–0.17	0.12–0.15
a	Lift-curve slope [1/rad]	5.5–5.75	5.5–5.75	5.5–5.75

The lift-curve slope sits near $2\pi \approx 6.28$ in theory, reduced to roughly 5.7 by finite-aspect-ratio and tip-loss effects, and falls further at very low Reynolds ($Re < 50\,000$).

7.2 Aerodynamic Trim Values

Symbol	Quantity	420 mm	550 mm	700 mm
$\bar{T}$	Hover thrust [N]	0.3–4	1–12	3–24
$\bar{v}_i$	Hover induced vel. [m/s]	2–8	3–11	4–13
$\bar{\theta}$	Trim pitch [deg]	4–10	5–12	5–12
τ_i	Inflow time constant [ms]	2–8	2–7	2–7

7.3 Derived Coefficients

Using $\rho = 1.225\,\mathrm{kg\,m^{-3}}$:

Symbol	Quantity	420 mm	550 mm	700 mm
A_t	Disc area [$\mathrm{m^2}$]	0.020–0.031	0.031–0.045	0.057–0.066
K_1	Pitch gain [N/rad]	25–150	50–180	130–275
K_2	Inflow gain [N s/m]	0.4–1.3	0.8–1.7	1.6–2.6
k_v	Inflow sensitivity [m/(s N)]	1.0–3.7	0.5–1.6	0.25–0.75
$K_2 k_v$	DC softening factor	0.8–2.8	0.6–1.9	0.5–1.5

The dimensionless product $K_2 k_v$ is in the range 0.5–3 across all three classes at typical operating points—a near-similarity result, since both K_2 and k_v scale with comparable combinations of ρA_t, $\Omega_t R_t$, and $\bar{v}_i$. Smaller helicopters tend to operate at lower thrust per unit disc area (λ_h smaller), pushing $K_2 k_v = \sigma a/(16\lambda_h)$ toward the upper end of the range.

7.4 Worked Example: 700 mm-class

Representative values for the tail rotor of a 700 mm-class electric helicopter (e.g. a large 3D/sport machine with main rotor radius ~ 0.79 m and total flying mass ~ 5 kg):

$$
\begin{aligned}
R_t &= 0.140 \text{ m}, & c &= 0.030 \text{ m}, & N_b &= 2, \\
\Omega_t &= 1036 \text{ rad/s}, & \Omega_t R_t &= 145 \text{ m/s}, & & \\
\sigma &= \tfrac{2 \cdot 0.030}{\pi \cdot 0.140} = 0.136, & a &= 5.7, & & \\
\bar{T} &= 12 \text{ N}, & A_t &= 0.0616 \text{ m}^2. & &
\end{aligned}
$$

Hence

$$
\begin{aligned}
\bar{v}_i &= \sqrt{12/(2 \cdot 1.225 \cdot 0.0616)} = 8.92 \text{ m/s}, \\
\tau_i &= 0.849 \cdot 0.140/(4 \cdot 8.92) \approx 3.3 \text{ ms (Pitt–Peters)}, \\
K_1 &= \tfrac{1.225 \cdot 0.0616 \cdot 145^2 \cdot 0.136 \cdot 5.7}{6} = 206 \text{ N/rad}, \\
K_2 &= \tfrac{1.225 \cdot 0.0616 \cdot 145 \cdot 0.136 \cdot 5.7}{4} = 2.13 \text{ N s/m}, \\
k_v &= 1/(4 \cdot 1.225 \cdot 0.0616 \cdot 8.92) = 0.372 \text{ m/(s N)}, \\
\lambda_h &= \bar{v}_i/(\Omega_t R_t) = 8.92/145 = 0.0615, \\
K_2 k_v\big|_{\text{consistent}} &= \frac{\sigma a}{16 \lambda_h} = \frac{0.778}{0.984} = 0.79.
\end{aligned}
$$

The DC softening factor is $S = 1/(1 + K_2 k_v) = 0.56$: the steady-state thrust delivered for a given collective is about 56% of the open-loop (frozen-wake) value.

7.5 Caveats

Pitt–Peters time constants are systematically low compared to flight-test data on full-scale rotors (factor of 2–4 underestimate); RC practitioners often use $\tau_i \approx 0.1$–0.3 rotor revolutions empirically. For design, the Pitt–Peters value should therefore be treated as a lower bound on the inflow lag, and either inflated by a fitted multiplier or verified against bench data before being used to set compensator coefficients. Blade lift-curve slope drops noticeably at low Reynolds numbers ($Re \sim 30\,000$–$80\,000$) typical of RC tail-rotor tips, especially with thin symmetric sections. Solidity calculations assume rectangular blades; tapered or paddle blades need an equivalent solidity weighted by $r^2 c(r)$. Finally, the tail rotor sits in the main-rotor wake, so $V_\perp$ in (21) is dominated by main-rotor inflow—the main source of tail-rotor effectiveness loss in turns.

8 Summary

The tail-rotor inner loop reduces to:

1. A linear blade-element thrust law (at fixed Ω_t), $T_t = K_1\theta_t - K_2(v_i + V_\infty)$, with constants from (16).
2. A first-order inflow state, $\tau_i \dot{v}_i + v_i = v_{i,\mathrm{ss}}$, driven by the momentum-theory steady-state map (21) and with τ_i from (27).
3. A linearised plant transfer function (44) about hover trim, exposing the lead–lag structure of the response and the DC softening factor $1 + K_2 k_v$.
4. A self-consistent closed form for the DC softening factor, $K_2 k_v|_{\mathrm{consistent}} = \sigma a/(16\lambda_h)$ (eq. 54), valid across the operating envelope.
5. An implementation-ready nonlinear dynamic inverse for the inner loop, (59)–(62), taking T_t^{cmd} and the flight condition $(V_\infty, V_\perp)$ as inputs and producing the servo pitch command θ_t, with the induced velocity v_i carried internally as a single state.